CSCI 441 SOFTWARE ENGINEERING · DEMO 1

Individual Contributions

Contribution breakdown and project management for Demo 1

Document type	**Individual contributions breakdown and project management report**
Project title	**SuperAsk.AI: answers about Cambodian government services, in Khmer and English, with the official source attached**
Group	**Team AO3**
Members	**Chun Phallamonunhh** **Fong Yean** **Sarin Sereyprophea** **Heng Bunkheang** **Choi Heejae** **Neang Doungkhe**
Date	**9 November 2026**
Repository	**https://github.com/Monunhh/Super-ask**

Evidence for each item is in the repository history and in Reports 1 and 2 (`docs/`).

Contents

1. Contribution summary 3

2. Contributions by item 3

3. Contributions by member 4

4. Demo presentation by use case 6

5. Items that do not apply and archive layout 7

6. Project management 7

- 6.1 Coordination and progress 7
- 6.2 Plan of work 8
- 6.3 Team responsibilities 8
- 6.4 Gantt chart 9

1. Contribution summary

Each member's contributions to Demo 1 are added up into one total percentage. All six members have reviewed this breakdown and agree with it.

Member	Role	Share of Demo 1
Chun Phallamonunhh	AI/ML engineer and system design	18%
Fong Yean	Project lead and manager	18%
Sarin Sereyprophea	QA and testing lead	15%
Heng Bunkheang	Web/API developer	18%
Choi Heejae	UX/UI designer	16%
Neang Doungkhe	Data and content curator	15%
Total		**100%**

2. Contributions by item

The items follow the contribution list in the Demo 1 requirements. ● marks the member who led the item; ○ marks a member who contributed to it.

Item	Chun	Fong	Sarin	Heng	Choi	Neang
1. Program code	●			●		
2. Unit and integration testing, coverage	●		○	○	●	
3. Front-end design	○			●	●	
4. Integration and cross-platform (web, Telegram)	○			●	○	
5. Testing of algorithms and requirements	●			○		
6. Debugging	●			○		
7. Program documentation	●				○	
8. Database design	Does not apply (see section 5)					
9. Data collection and scripts	●			○		

Item	Chun	Fong	Sarin	Heng	Choi	Neang
10. Brochure	○	●				
11. Slides	●			●		
12. Project management	○	●	○			○
Demo presentation	○	○	●	○	○	●

3. Contributions by member

Chun Phallamonunhh

18%

- Built the Phase 0 prototype: the three-tier answer engine, guardrails, retrieval and ranking, Khmer normalisation and segmentation, claim verification, citations, model providers and the extractive fallback, and the audit log with personal-data redaction.
- Wrote the automated test suite (610 unit tests, 33 API route integration tests, 89.6% line coverage), the retrieval and refusal evaluation, the quality-gate scripts and the GitHub Actions CI workflow.
- Built the source registry and the crawler that monitors 132 government websites, and the knowledge-base format with provenance and dates.
- Wrote the use case descriptions, use case diagram and fully dressed use cases (Report 1), and the interaction diagrams, class specifications and system architecture revisions (Report 2).
- Organised the repository and the Demo 1 e-archive, wrote README1–4 and rebuilt the slides to five.
- Wrote the technical documentation, user documentation and this individual contributions report.
- Opens the demo and presents UC-1 Ask a question and UC-2 Retrieve official sources.

Evidence: commits by Chun Phallamonunhh in the repository history (earlier ones under the git name "Your Name" with the same email), from the initial prototype on 31 July 2026 onward; `lib/`, `tests/`, `scripts/`, `.github/workflows/ci.yml`.

Fong Yean 18%

- Managed the project: studied each deliverable's requirements, broke the work into tasks, assigned them to team members and followed up until they were done.
- Was the team's main point of contact with the professor, and handled team communication and the submission of each deliverable.
- Wrote the goals, system requirements and analysis, and contributed to actors and goals (Report 1).
- Combined and formatted Report 1, captured prototype screenshots, and prepared the project management section: Gantt chart, team responsibilities and references.
- Wrote the algorithms and data structures, user interface design and implementation, and design of tests sections (Report 2).
- Coordinated the Demo 1 submission: prepared the brochure and uploaded the integration test files to the e-archive.
- Kept the plan of work and schedule up to date and checked consistency across the team's documents.
- Presents the plan for the rest of the semester.

Evidence: commits by YeanFong on 9 November 2026 (documents, brochure, integration test upload); Reports 1 and 2, sections listed above.

Sarin Sereyprophea 15%

- Wrote the customer problem statement and the stakeholders section, and reviewed requirements and QA-related content (Report 1).
- Updated the project plan, milestones, deadlines and schedule (Report 2).
- Presents UC-4 Decline an unsupported question.

Evidence: Reports 1 and 2, sections listed above.

Heng Bunkheang 18%

- Added the Telegram bot and the Telegram Mini App channel, and rebranded the prototype to SuperAsk.
- Implemented the site redesign from the SuperAsk Figma design (landing page, chat, colours) and fixed the CI coverage configuration.
- Built the first HTML version of the Demo 1 slides with its PDF export.
- Prepared the traceability matrix and system sequence diagrams (Report 1), and the subsystems, package diagrams, architectural styles and hardware mapping (Report 2).
- Presents UC-7 Office contact and location, in Telegram.

Evidence: commits by bunkheangheng / HENG Bunkheang, 16 September to 8 November 2026, including pull request #1 (Figma redesign); `lib/telegram/`, `components/`.

Choi Heejae 16%

- Designed the user interface and user experience, and wrote the preliminary design and user effort estimation sections (Report 1).
- Documented system connectors, network protocols and other architecture specifications (Report 2).
- Wrote the 9 cross-channel integration tests (web Q&A and citations, Khmer and English, declines, feedback, Telegram) with their configuration and README3.
- Presents UC-3 View the source citation.

Evidence: commit by CHOI HeeJae on 8 November 2026 adding the integration tests; `docs/demo–1/3_integration_testing/`.

Neang Doungkhe 15%

- Researched Cambodian government-service information and wrote the problem breakdown, glossary, and actors and goals sections (Report 1).
- Coordinated team responsibilities, gathered contribution details and checked consistency between parts of Report 2.
- Presents UC-11 Refresh official content.

Evidence: Reports 1 and 2, sections listed above.

4. Demo presentation by use case

The demo is organised by use case, and each use case is presented by the member who owns that area (Report 1, product ownership and team responsibilities).

#	Part	Business goal	Presenter	Time
1	Overview and roadmap		Chun Phallamonunhh	2 min
2	UC-1 Ask a question, UC-2 Retrieve official sources	Bilingual access, comprehensive information	Chun Phallamonunhh	4 min
3	UC-3 View the source citation	Trustworthy answers	Choi Heejae	2 min
4	UC-4 Decline an unsupported question	Trustworthy answers	Sarin Sereyprophea	2 min
5	UC-7 Office contact and location, in Telegram	Ministry contact and location	Heng Bunkheang	3 min
6	UC-11 Refresh official content	Comprehensive, current information	Neang Doungkhe	2 min

#	Part	Business goal	Presenter	Time
7	Plan for the rest of the semester		Fong Yean	2 min

5. Items that do not apply and archive layout

Code, unit tests and data collection scripts stay at the repository root

The folders `1_code`, `2_unit_testing` and `4_data_collection` in the e-archive hold a README each (README1, README2, README4) instead of a copy of the files. SuperAsk is a Next.js application, and Next.js requires its source folders (`app/`, `components/`, `lib/`) at the repository root. The unit tests in `tests/` and the data collection scripts in `scripts/` and `data/registry/` import that code and read and write `data/` by path, so they must stay there too. A copy in the e-archive would be a second version that goes out of date. Each README gives the exact location and the commands to run the code. The integration tests have their own configuration and are kept in `3_integration_testing`.

Nothing is missing, so we do not propose moving any points away from items 1, 2 and 9.

Item 8, database design, does not apply

SuperAsk has no database in Phase 0. The knowledge base, curated answers, glossary, office directory and source registry are version-controlled JSON files in `data/`, so every change to official content is reviewed and recorded in git. Audit and feedback records are append-only log files. We propose moving the points for item 8 to item 9, data collection, which covers the design of those data files and the registry of 132 government websites.

6. Project management

6.1 Coordination and progress

For Demo 1 the team delivered a working prototype with web chat, a Telegram bot and Mini App, Khmer and English support, source citations and declines, backed by 652 automated tests and a 44-question evaluation. The demo shows the use cases planned for Demo 1 in Report 1 (UC-1 to UC-4, UC-7 and UC-11), each presented by its owner.

Fong Yean managed the project: she read the requirements for each deliverable, assigned the work to the team, communicated with the professor, and coordinated the schedule, the reports and the submissions. Chun Phallamonunhh maintained the developer resources: the GitHub repository, the CI workflow, the Vercel deployment and the e-archive layout. Sarin Sereyprophea tracked milestones and deadlines; Neang Doungkhe gathered contribution details. Each member prepared the use case they present.

6.2 Plan of work

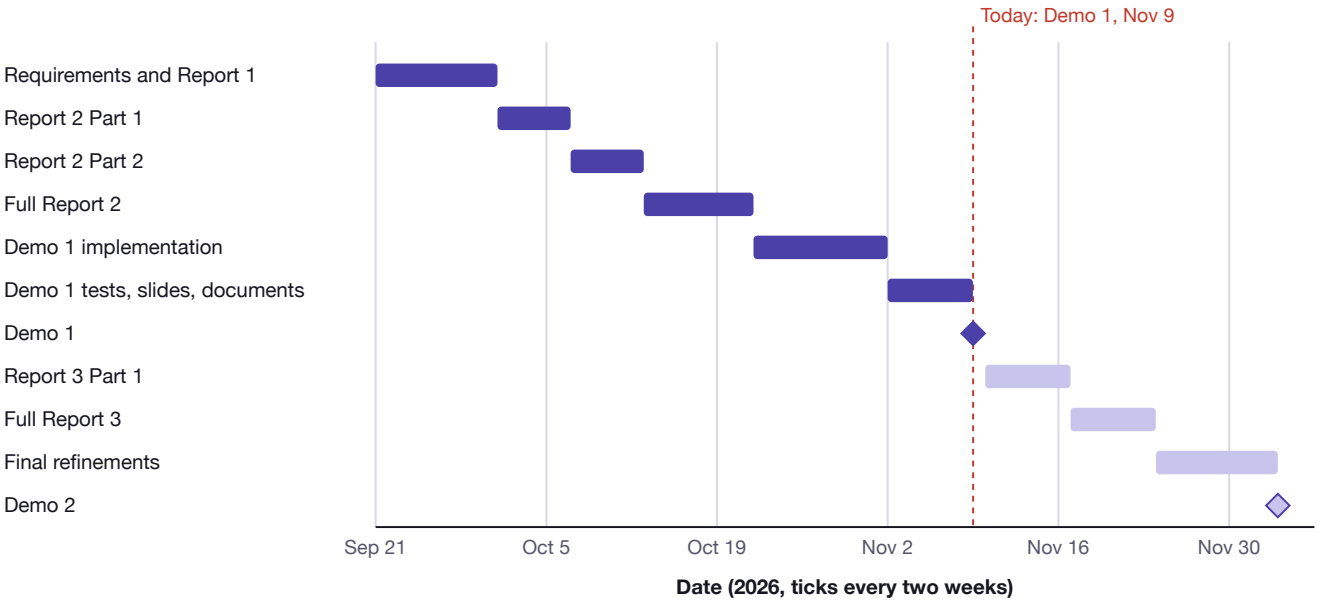
Period	Planned work	Milestone
Oct 22 – Nov 1	Develop the Demo 1 use cases; write tests and documentation	Demo 1 implementation
Nov 2 – 8	Test the web and Telegram prototype, evaluate retrieval and declines, prepare slides, brochure and e-archive	Demo 1 preparation
Nov 9	Present the working prototype; submit the e-archive	Demo 1
Nov 10 – 16	Apply Demo 1 feedback: sharper relevance check so uncovered topics are declined	Report 3 Part 1 (Nov 16)
Nov 17 – 23	Add approved content for passports and business registration; improve ranking (right source first, 67.6% today)	Full Report 3 (Nov 23)
Nov 24 – Dec 3	Office directory with map links (UC-7), Telegram follow-up memory, real user accounts, reviewer portal for monitored changes	Final refinements
Dec 4	Present the final system	Demo 2

6.3 Team responsibilities

Member	Role	Demo 1	After Demo 1
Chun Phallamonunhh	AI/ML engineer and system design	Answer engine, retrieval, tests, CI, repository and e-archive; opens the demo and presents UC-1 and UC-2	Improve ranking and relevance; reviewer portal; keep technical documents current
Fong Yean	Project lead and manager	Requirements and task assignment, communication with the professor, schedule, reports, brochure, submission; presents the plan	Coordinate Report 3 and Demo 2 documents; track deadlines
Sarin Sereyprophea	QA and testing lead	Reviews requirements and QA content; presents UC-4	Usability checks; verify fixes; coordinate final demo testing
Heng Bunkheang	Web/API developer	Telegram bot and Mini App, site redesign; presents UC-7	Real user accounts; office directory with map links; Telegram memory

Member	Role	Demo 1	After Demo 1
Choi Heejae	UX/UI designer	UI design, integration tests; presents UC-3	Refine the bilingual, responsive interface from feedback; integration checks
Neang Doungkhe	Data and content curator	Government-service research; presents UC-11	Add and check content for passports and business registration

6.4 Gantt chart



Project schedule from Report 1 to Demo 2. Horizontal axis: calendar date in 2026. Dark bars: completed work; light bars: planned work; diamonds: demos. The dashed line marks today, 9 November 2026.